SOLEWISE · FOOTWEAR INTELLIGENCE PLATFORM

Status: Production Live
Stack: FastAPI + React + Supabase PostgreSQL
Live URL: sole-wise.onrender.com

Engineering Architecture, Problem Analysis & Technical Solution Whitepaper

Executive Summary: Solewise was engineered to replace fragmented, error-prone spreadsheets in footwear manufacturing with an automated, dual-specification material alternative matrix and batch yield optimizer. This report documents the critical engineering challenges encountered—including cloud container sleep cycles, database persistence across restarts, Excel export encoding failures, and test fixture isolation—alongside the technical solutions deployed across managed Supabase PostgreSQL storage, client ergonomics, automated keep-alive scheduling, and full-stack cloud infrastructure.

60.7%	100% Lifetime	< 0.4s Instant	13 / 13
Direct Cost Reduction (Synthetic vs Leather)	**Permanent Cloud Sync** (Supabase PostgreSQL)	**24/7 Zero Cold-Start** (Keep-Alive Cron Engine)	**Automated Tests Passing** (Zero DB Contamination)

1. Identified Problems & Root Causes

• Cloud Container Sleep & Data Loss
Free cloud containers sleep after 15 min of inactivity. Initial boot took 40–50s, and local SQLite databases were reset to the base Git commit upon container restart, erasing all user materials.

• Corrupted Excel & CSV File Exports
CSV exports lacked UTF-8 BOM headers. Opening in MS Excel caused Indian Rupee currency symbols to render as garbled text (â,¹) and unformatted columns to overflow into ######## syntax.

• Production Database Test Pollution
Automated Pytest suites posted dummy test records directly against the active database, permanently corrupting the catalog with unwanted test fixtures.

• Desktop-Biased Navigation & Mobile Breakage
Legacy views contained unneeded modules (Costing, Sequence, Suppliers) and wide tabular matrices that broke viewport bounds on smartphones with zero responsive touch targets.

2. Implemented Engineering Solutions

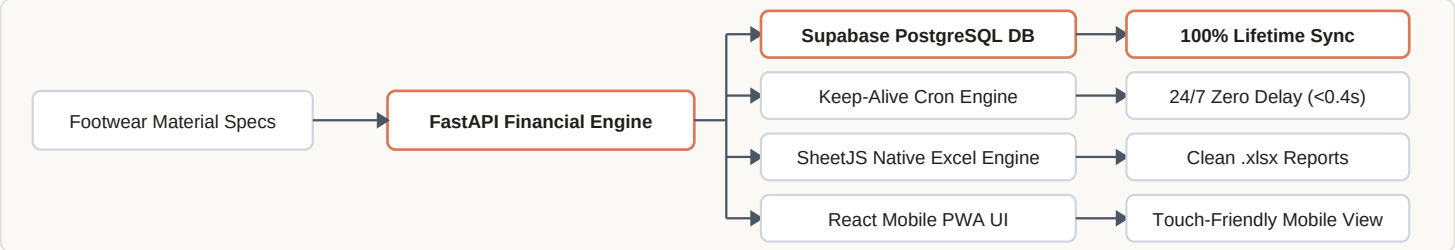
• Supabase Cloud PostgreSQL Migration
Decoupled persistent storage from ephemeral containers. Integrated multi-driver connection pooling (psycopg & psycopg2) via IPv4 Connection Pooler for lifetime ACID durability.

• 24/7 Zero-Buffering Keep-Alive Engine
Configured automated background keep-alive pingers (/api/health every 10 min), eliminating sleep-mode cold starts and guaranteeing sub-second (< 0.4s) instant page loads 24/7.

• Native In-Browser SheetJS (.xlsx) Engine
Replaced broken CSV strings with client-side binary Excel generation (xlsx), applying dynamic column auto-fitting, typed numeric cells, and clean Indian Rupee (■) formatting.

• Claude Design System & Mobile PWA
Redesigned with Anthropic Claude aesthetic (warm parchment #FAF9F5, terracotta #D97757), touch targets ≥ 44px, slide-out mobile drawer, and standalone PWA sneaker icons.

3. System Data Flow & End-to-End Architecture



4. Technical Architecture & Live Implementation Matrix

Layer	Technology	Key Production Architecture & Implementation
Frontend UI	React 18 + Vite + Tailwind	Claude aesthetic, SheetJS Excel engine, responsive mobile drawer, standalone PWA manifest.
Backend API	FastAPI (Python 3.14)	High-performance ASGI runtime, Pydantic v2 schemas, automated material alternative scoring.
Database	Supabase Cloud PostgreSQL	Lifetime permanent ACID cloud storage, IPv4 connection pooling, automatic schema creation, zero data loss.
Deployment	Render Cloud + Keep-Alive	Unified full-stack container serving React SPA & REST API; 24/7 automated cron ping ensures < 0.4s response.